

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/376529851>

Optimization Of Genetic Algorithm Using PSO In Food Menu Recommendation System

Conference Paper · November 2023

DOI: 10.1109/ICIMCIS60089.2023.10348618

CITATIONS

0

READS

109

6 authors, including:



[Ria Andriani](#)

University of Indonesia

10 PUBLICATIONS 18 CITATIONS

[SEE PROFILE](#)



[Ainul Yaqin](#)

Universitas Amikom Yogyakarta

70 PUBLICATIONS 241 CITATIONS

[SEE PROFILE](#)



[Kraugusteeliana Kraugusteeliana](#)

Universitas Pembangunan Nasional Veteran Jakarta

58 PUBLICATIONS 254 CITATIONS

[SEE PROFILE](#)



[Intan Hesti](#)

Universitas Pembangunan Nasional Veteran Jakarta

9 PUBLICATIONS 18 CITATIONS

[SEE PROFILE](#)

Optimization Of Genetic Algorithm Using PSO In Food Menu Recommendation System

Ria Andriani

Faculty of Computer Science,
Universitas Amikom Yogyakarta
Jl Ringroad Utara, Condongcatur,
Depok, Sleman, Yogyakarta Indonesia
55283

ria@amikom.ac.id

Ade Pujianto

Faculty of Computer Science,
Universitas Amikom Yogyakarta
Jl Ringroad Utara, Condongcatur,
Depok, Sleman, Yogyakarta Indonesia
55283

ade.pujianto@amikom.ac.id

Ainul Yaqin

Faculty of Computer Science,
Universitas Amikom Yogyakarta
Jl Ringroad Utara, Condongcatur,
Depok, Sleman, Yogyakarta Indonesia
55283

ainulyaqin@amikom.ac.id

Hanif Al Fatta

Faculty of Computer Science,
Universitas Amikom Yogyakarta
Jl Ringroad Utara, Condongcatur,
Depok, Sleman, Yogyakarta Indonesia
55283

hanif.a@amikom.ac.id

Kraugusteeliana Kraugusteeliana

Faculty of Computer Science,
Universitas Pembangunan Nasional
Veteran Jakarta
Jl. RS. Fatmawati, Pondok Labu,
Jakarta Selatan, 12450

kraugusteeliana@upnvj.ac.id

Intan Hesti Indriana

Faculty of Computer Science,
Universitas Pembangunan Nasional
Veteran Jakarta
Jl. RS. Fatmawati, Pondok Labu,
Jakarta Selatan, 12450

hesti@upnvj.ac.id

Abstract — Covid-19 is a pandemic caused by a virus and attacks the whole world, one way to overcome this is by regulating diet, but not everyone is able or understands how to maintain a healthy lifestyle. The lack of knowledge about foods rich in protein, nutrients, minerals and vitamins extends the healing process of Covid-19 by increasing the immune system. Therefore we need a system with artificial intelligence (AI) that can provide food menu recommendations for the general public. The menu can sThe genetic algorithm is used to carry out training data and provide prediction and decisions based on existing data, and the PSO algorithm is used to optimize the learning process of the genetic algorithm, the algorithm optimization is carried out with a configuration of max population 200, max generations 500, mutation rate 0.03, this configuration produces the best value with a time of 2284ms, iterations 309 and accuracy 90.58%

Keywords— Covid-19, Genetic, PSO, AI, Food menu

I. INTRODUCTION

Coronavirus disease 2019 or Covid-19 is a new disease that is happening all over the world. Covid-19 first appeared in the city of Wuhan, China, in December 2019. This virus has a fast spreading pattern so that it is declared an epidemic and causes a global pandemic. Since the beginning of its spread, Covid-19 has caused a high death rate. Currently, it has affected 36.5 million people and caused 1.06 million deaths worldwide. In Indonesia, until now there have been 325,000 confirmed positive cases with a total of 11,677 deaths and 359,000 recovered [1].

Specific drugs have yet to be found to treat the disease caused by the virus. One of the most effective ways is to do a vaccine against Covid-19, but in reality, there are still people who are affected by Covid-19 even though they have done the vaccine. This can happen because the human immune system will adapt to the Covid-19 virus. However, the human immune system varies based on the age and health conditions of the individual. Therefore, it is necessary to carry out a series of activities to boost the immune system and fight the Covid-19 virus effectively [2].

The most effective way to increase the human body's resistance is through physical exercise and meeting nutritional needs for protein, nutrients, minerals and vitamins. In some cases, especially in Europe, many sports athletes such as football players have been affected by the Covid-19 virus. However, on average, these athletes recovered within days of

initially being reactive to the Covid-19 virus. This is because when athletes test positive, they follow health protocols such as self-isolation and practicing social distancing. In addition to following health protocols, these athletes also carry out activities to increase endurance through a healthy lifestyle, including regular exercise and consuming foods rich in protein, nutrients, minerals and vitamins [3].

However, not everyone is able or understands how to maintain a healthy lifestyle. The lack of knowledge about foods rich in protein, nutrients, minerals and vitamins extends the healing process of Covid-19 by increasing the immune system. The increasing number of patients affected by the Covid-19 virus also burdens the health team, such as doctors and other medical volunteers, because they have to treat Covid-19 patients. Therefore, a system is needed to help and provide education to the public, especially those affected by Covid-19. Through this system, it is hoped that it can help the community by providing education and recommendations for choosing food to increase endurance and recover from Covid-19 [4].

An artificial intelligence (AI) method is needed to provide accurate recommendations or decisions, where the method can provide reasons and predictions based on existing data. One of the well-known AI algorithms for inferring is the genetic algorithm, which can reason through mechanisms such as DNA connections to get the best genetic mutations. To try to improve the performance of the genetic algorithm, researchers tried to optimize it by adding the Particle Swarm Optimization (PSO) algorithm. The PSO algorithm is the most widely used and the most optimal when compared to other algorithms such as CSO (Cat Swarm Optimization), LSA (Lightning Search Algorithm), CSA (Crow Search Algorithm), WOA (Whale Optimization Algorithm), SCA (Sine Cosine Algorithm), WDO (Wind Driven Optimization) and GSA (Gravitational Search Algorithm) [1]. The aim is to optimize the reasoning process in the genetic algorithm through several mechanisms and configurations in the two algorithms [5].

II. STUDY OF LITERATURE

In this research, a new hybrid algorithm named GA-HIDMS-PSO is introduced. This algorithm combines advanced and dynamic PSO (Particle Swarm Optimization), namely Heterogeneous Enhanced Multi-Swarm PSO (HIDMS-PSO), with Genetic Algorithm (GA). This new

hybrid model combines the heterogeneous features of HIDMS-PSO and the evolutionary characteristics of GA. In the GA-HIDMS-PSO architecture, HIDMS-PSO acts as the primary search engine, while GA is used as a secondary method that helps maintain homogeneous and heterogeneous subpopulations of the HIDMS-PSO algorithm. These two methods run sequentially, with HIDMS-PSO running longer than GA. HIDMS-PSO provides initial solutions for GA from homogeneous and heterogeneous subpopulations, while the final solutions returned from GA are used to replace previous solutions in HIDMS-PSO. This allows the HIDMS-PSO search process to involve a wider variety of particle types, thereby better guiding the swarm. The performance of the GA-HIDMS-PSO algorithm was tested in two test sets, namely the CEC'05 and CEC'17 test series, with dimensions 30 and 50. The test results showed that the GA-HIDMS-PSO outperformed all 24 algorithm comparisons, including 12 advanced PSO variants, and 12 other metaheuristics, on both sets of tests for both dimensions [5].

The research presents a scheduling strategy for future Energy Storage Systems (ESS) in microgrids. This strategy involves the use of two algorithms, namely Genetic Algorithm (GA) and Particle Swarm Optimization (PSO). The aim of this scheduling strategy is to minimize costs that must be borne by consumers in microgrids that experience dynamic price changes. then formulates the objective function used in the optimization, describes the associated search space, and the nature of its structural analysis. In this study, we prove that the search space has a minimum limitation of $50 \times (Bc - Bd + 1)$, with Bc and Bd representing the maximum charge and discharge rates in one hour (in proportion) on the ESS. Through simulations involving load prediction, energy generation, and grid pricing for three microgrids of different sizes, we managed to derive an ESS Schedule that can reduce costs by an average of 11.31% (using GA) and 14.31% (using PSO).) when the ESS Schedule is implemented using the Clean Power Based Algorithm [6].

This research aims to optimize the scheduling process, it is important to use the method that produces the best optimal results. In this paper, we will compare the effectiveness of Genetic Algorithms and Particle Swarm Optimization in designing class schedules. After implementing the two algorithms, we conducted a comparative analysis by assessing the fitness value and execution speed of each algorithm during the lecture scheduling process. The results showed that in the 25th iteration, the Genetic Algorithm succeeded in producing a lecture schedule with the best match value of 0.021 and an execution time of around 9.36 seconds. While the PSO algorithm produces a match value of 0.099 and an execution time of around 61.95 seconds in the same iteration. These results indicate that the PSO algorithm is superior in terms of match value compared to the Genetic Algorithm. However, it should be noted that the Genetic Algorithm has a faster execution time compared to the PSO algorithm [7].

In this study, we tried to use a Genetic Algorithm to optimize the Random Forest model to get the best parameters and then apply it to the heart failure dataset from Kaggle. after experimenting with two iterations for every nine parameter combinations. The results of the optimized Random Forest model are compared with the Random Forest without optimization, decision trees and the Naïve Bayes algorithm.

Our findings show that the optimized model performs slightly better than other algorithms. The best F1 score was obtained in the second iteration with a value of 0.90789, while Random Forest without optimization obtained a score of 0.89404, decision tree 0.85034, and Naïve Bayes 0.86195. The best recall value is 0.91925 obtained in the first and second iterations. Whereas in terms of the best precision with a value of 0.89937 achieved in the first iteration. With these results, the optimized Random Forest algorithm can be used to generate reliable predictions about heart failure [8].

In this study, food composition was prepared by taking into account the amount of nutrition, salt content, and minimum costs. The proposed method is a hybrid Genetic Algorithm (GA) and Variable Neighborhood Search (VNS) method. Three GA-VNS hybrid type scenarios have been developed in this study. Even though hybrid GA-VNS takes more time than pure GA or pure VNS, the proposed method provides better solution quality. VNS managed to help GA avoid premature convergence and raise better solutions. Weaknesses of GA in local exploitation and premature convergence are overcome by VNS, while deficiencies in VNS which lack global exploration capabilities can be overcome by using GA which has advantages in global exploration [9] .

This study aims to regulate daily food consumption patterns based on the Healthy Food Guidelines (H4P5) or the Healthy 4 Perfect 5 principle. The Healthy Food Guidelines do not provide guidelines regarding portions and sizes for daily food consumption, therefore these Guidelines have been revised to become Balanced Nutrition Guidelines (GBN). The GBN regulates consumption patterns consisting of three servings of protein vegetables, three servings of animal protein, eight servings of carbohydrates, five servings of vegetables, five servings of fruit and eight servings of water. Each portion has a size and dose according to the GBN principle. To determine the right portion, optimization is used using Genetic Algorithm (GA) on food composition. Optimization with GA produces three outputs. First, a comparison is made between the value of the initial objective function (0) and the value of the generative objective function (1), with a ratio of 6.13:4.72. The smaller this ratio, the better the quality of the chromosomes. Second, 50% mutation probability, 5% crossover probability, maximum number of generations is 36, maximum fitness point is 0.240806, and population is 6. Third, optimization using Genetic Algorithms on food composition produces one food model, one consumption model, and one composition model. Thus, the results of this study (food models, consumption models, and composition models) can be used as recommendations for food consumption patterns in normal circumstances and as guidelines for national food consumption policies [10].

In this research, the integration of fuzzy evaluation and genetic operators into Particle Swarm Optimization (PSO) was carried out with the aim of increasing decision-making abilities and increasing the possibility of selecting solutions that have higher quality. In the research results it is proven that by evaluating fuzzy and genetic operators into classical PSO, the limitations in producing less than optimal solutions can be successfully overcome. This finding has a very important meaning in the decision-making process that aims to significantly improve the quality of dietary meal planning [11].

This research introduces a new hybrid algorithm that combines Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) called PSOGA. The goal is to get the optimal truss structure design with discrete design variables. The objective function chosen in this study is the total weight of the frame structure, which depends on the specified stress limits and displacement limits. PSO basically models social behavior of birds, in which individual birds exchange information about position, speed and fitness, and influence flock behavior to seek other areas of high fitness. However, PSO tends to get stuck at local points because the movement is not uniform. To overcome this, this study uses natural genetic operators such as mutation, random selection, and reproduction in GA to achieve the best genetic algorithm. In this PSOGA algorithm, identical chromosomes or particles can meet and interact with each other, helping in finding the optimal design through deterministic and probabilistic rules. This study conducted experiments by designing several reference truss structures using the proposed new algorithm, and the results were compared with other evolutionary optimization methods [12].

This research was conducted by applying the Genetic Algebra Method to arrange class schedules and practicums. The Genetic Algebraic Method is an algorithm inspired by evolutionary principles in biology. The main objective of this research is to develop a computerized system that can replace manual lecture and practicum scheduling methods. By using the Genetic Algorithm, it is hoped that the resulting schedule will be more accurate by following predetermined rules, and can be completed in a shorter time. Lecture scheduling data remains the basis for applying this method, and the results of using genetic algorithms can be recommended for scheduling lectures efficiently and according to the generated schedule [13].

This research attempts to arrange teachers' teaching schedules in schools which is a complex task. There are various aspects involved in scheduling, such as synchronizing the teacher's schedule with other class schedules, as well as the distribution of teaching and learning activities so that the schedule for each class is evenly distributed every day. Therefore, in this research an application will be developed to arrange class schedules automatically using a genetic algorithm. This application aims to simplify and speed up the process of preparing course schedules at SMKN 1 Malang by considering existing resources and limitations. It is hoped that this application will also improve communication between teachers and students in managing lesson schedules, thereby achieving optimal results for all parties involved [14].

This study concludes that the genetic algorithm works by changing the relationship structure and rearranging the table to find the best solution, which is determined based on the fitness value. The highest fitness value of each experimental result is used. For this experiment, we used a MySQL sample database named "employee" with a total of more than 3,000,000 rows distributed in 6 tables. The query is designed using 5 relations in the form of a left deep tree. Before optimization, the query execution time was 8.14247 seconds. After implementing genetic algorithm optimization, the execution time was reduced to 6.08535 seconds, resulting in a 25.3% time reduction. This shows that the genetic algorithm effectively reduces query execution time by optimizing the relation part. Thus, the use of genetic algorithms to optimize

queries can be an alternative solution to maximize query performance [15].

The results of this study show that the basic algorithm, namely Cloud Particle Swarm Optimization (CPSO) in 500 generations has not reached convergence 8 times, 6 times, 4 times, and 5 times, respectively. In the case of convergence, the average number of steps is much higher than that of ICPSO (Improved Cloud Particle Swarm Optimization), and the improved algorithm completely converges. In terms of time performance, the ICPSO convergence time is much better than the CPSO algorithm. Therefore, the improved particle optimization algorithm guarantees the effectiveness of the improvement steps, such as small optimization algebra, fast combining speed, high efficiency and good population diversity [16].

This study examines the generalizability of two variants of the particle swarm optimization (PSO) algorithm, namely PSO with fixed phase (PSOFP) and PSO with dynamic phase (PSODP), which has been verified through experimental studies with six variants of the standard PSO algorithm on 12 benchmark functions for single-purpose numerical optimization. The studies cover 50 and 100 dimensions for each benchmark function, and the experimental results verify the generalizability of the proposed PSO variant based on the thesis building block concept [17].

This research discusses cloud computing and the importance of scheduling work in a cloud environment to maintain service quality and increase system efficiency. To overcome the problems of long scheduling times and high computation costs associated with job scheduling in the cloud environment, this research introduces a scheduling technique based on the optimization of a modified particle swarm. In this approach, the PSO algorithm is modified by adopting biological changes to avoid premature convergence and improve local search ability. Simulation results using CloudSim show that the proposed scheduling technique is effective in reducing free time and costs efficiently compared to standard PSO [18].

This research attempts to use the particle swarm optimization algorithm to optimize and analyze ship energy consumption based on ship energy consumption data during navigation, as well as test ship energy efficiency through the ship's energy efficiency operating index. The results of the study show that theoretically, the use of particle swarm optimization algorithms to optimize ship speed is a feasible step to reduce ship energy consumption and ultimately achieve the goals of energy saving and emission reduction [19].

This study aims to solve the portfolio optimization problem using the mean-semivariance (MSV) model. An adaptive particle swarm optimization multi-strategy algorithm named APSO/DU was developed to overcome this challenge. The use of constraint factors and the dual-update (DU) strategy improves the performance of the algorithm. The results show that APSO/DU has better accuracy and convergence speed and is closer to the global Pareto front (PF). This algorithm promises to provide valuable advice to investors and has potential practical applications in stock portfolio management, assisting in making smarter investment decisions and having a positive impact on investment returns [20].

This study aims to demonstrate the application of the Particle Swarm Optimization algorithm in determining material parameters for the isotropic incompressible elastic strain energy function using stress and torque loading. Rubber behavior simulation was carried out by modeling cylinder deformation using an isotropic hyperelastic incompressible material. In this study, four different forms of the strain-energy function are considered, including polynomial, exponential, and logarithmic terms, to reproduce the trend of load force (N) and torque (M) using natural rubber experimental data. The results show that the Particle Swarm Optimization algorithm succeeded in identifying the best material parameters and was able to characterize the behavior of hyperelastic materials such as rubber through the resulting objective function minimization process [21].

III. RESEARCH METHOD

This section contains a detailed and comprehensive research flowchart, including algorithms, rules, modeling, and related aspects of system design. The research flow follows the Application Development method which is commonly used by indie system developers. The stages involved in the indie development method are as follows (as shown in Figure 1):

a. Concept Development

The Concept Development stage is the most important step. At this stage, researchers will carry out Concept Development through mechanisms such as literature review, data and information collection, and definition of needs.

b. Design

This stage defines the detailed requirements of the system to be developed. The design phase also involves implementation analysis and optimization of the algorithms to be used.

c. Execution

At this stage, the code is written at the same time as the creation of the required assets. This step follows the Design phase, and involves implementing the system design and performing tests. Algorithmic optimization, such as Particle Swarm Optimization (PSO) and genetic algorithms, will be carried out in this process. Optimization is necessary because, during the mutation process of the genetic algorithm, genes in chromosomes may collide or cross each other, causing a search for new, old genes. To avoid collisions in the genetic algorithm, optimization is carried out during the training process using PSO.

d. Testing

Testing is carried out after the Execution stage to evaluate application functionality and assess the performance of algorithm optimization in system development. This research includes two test methods, namely Application Testing and Algorithm Testing. Application Testing involves Alpha Testing, which tests functionality before the system is released. Beta testing is carried out by related experts who evaluate the results of decision making from the system. Scenario Testing is also carried out to test the system through a certain mechanism. Algorithm testing or validation will be carried out using the Confusion Matrix and Cross Validation, which assess performance based on parameters such as accuracy, precision, recall, number of iterations, and training

time of the optimized artificial intelligence (AI) algorithm used.

e. Application

Implementation stage is the final stage of the research, in which the two algorithms have been implemented and tested to get the best combining mechanism in making food menu recommendations for people with Covid-19.

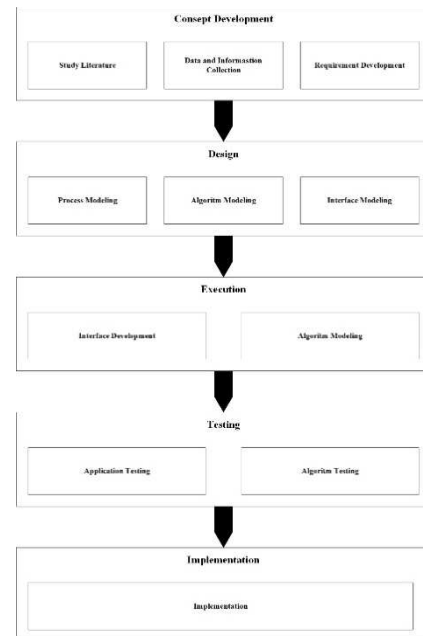


Fig. 1. Research Method

IV. RESULTS

In the Results and Analysis section, two testing mechanisms will be carried out: application/system testing and algorithm (AI) testing to assess the performance of the system being developed. The results and detailed research analysis are as follows:

a. System Design

At this stage it will be described through the development of a flow board, which is a graphical representation that describes a step by step solution to a problem. Flowcharts are used not only to illustrate simple operations but also to solve complex problems. The following is a flowboard of the Covid-19 patient menu recommendation system.

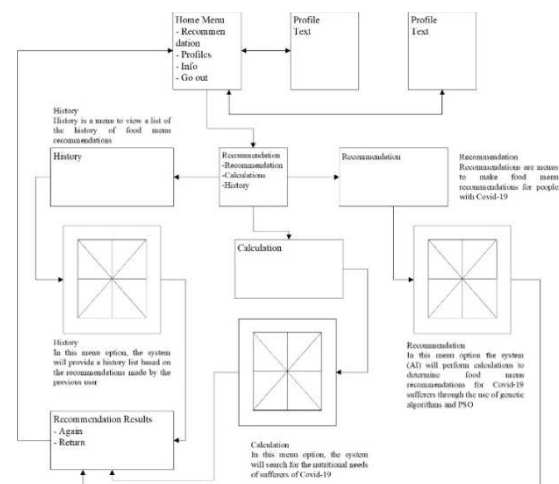


Fig. 2. Flowboard System

b. Application Testing

Application testing is carried out to assess whether the implementation of the algorithm in the application functions properly and correctly. Application testing in particular will focus on the core part of the system, namely the menu recommendation which is the center of algorithm impomentation. The recommendation menu consists of several stages, starting from the initial status to the final recommendation status. The initial state involves filling in personal data such as gender, age, height, weight and current health condition. This status will serve as input for the a algorithm to provide accurate recommendations. The test results are depicted in Fig. 3.

The screenshot shows three screens from the application. The first screen, 'Hasil Rekomendasi', lists three recommendations with their nutritional values. The second screen, 'Detail Rekomendasi 4', shows a detailed meal plan for a day (Makan Pagi, Makan Siang, Makan Malam) with specific food items and quantities. The third screen shows 'Rekomendasi 1', 'Rekomendasi 2', and 'Rekomendasi 3' with their respective nutritional data.

Fig. 3. Application Testing

c. Algorithm Testing

Algorithm testing was carried out on the genetic algorithm used in the food menu recommendation system for people with Covid-19 using the Violated Directed method for mutations, an extended intermediate method for crossover, and a random selection method for selecting parents. The selection method used is Fitness Based Selection with Elitism principle. In addition, particle swarm optimization is used to improve training and evaluation results. Testing is done with a dataset configuration consisting of 500 data points.



Fig. 4. Implementation of genetic algorithms and PSO in Applications

Fig. 4 illustrates the possible outcomes or patterns that will be executed by AI (computer). From these various possible results, AI (computer) will choose the best path in making decisions to make food menu recommendations.

Algorithm performance was tested through three testing mechanisms, namely test 1 with the particle swarm optimization (PSO) algorithm , test 2 with the genetic algorithm (GA) , and test 3 with the genetic algorithm optimized with the particle swarm optimization (PSOGA) algorithm . The detailed performance of the algorithm testing can be seen in TABLE I based on the Quick Testing mechanism.

TABLE I. THECONCLUSION

Scenarios	Configures	Times (ms)	iterations	accuracy
Scenarios -1 (PSO)	Max population = 200 Max generations = 500 Mutation rate = 0.03	1892	270	87.09%
Scenarios -2 (genetic)	Max population = 200 Max generations = 500 Mutation rate = 0.03	1520	202	88.20%
Scenarios -3 (PSOGA)	Max population = 200 Max generations = 500 Mutation rate = 0.03	2284	309	90.58%

CONCLUSION

Based on the results of the study it can be concluded that in the configuration of Max population 200, Max generation 500, and Mutation rate 0.03 with a dataset of 500 data, the following conclusions are obtained: In test 1 using the particle swarm optimization (PSO) algorithm, the average time is 1892ms, with 270 iterations and 87.09% accuracy . In test 2 using the genetic algorithm (GA) the average time is 1520ms, with 202 iterations and 88.09% accuracy. In test 3 using a genetic algorithm optimized with particle swarm optimization algorithm, the average time is 2284ms, with 309 iterations and 90.58% accuracy (PSOGA) . Based on these results, the combination of genetic algorithms and particle swarm optimization (PSOGA) can increase the accuracy and effectiveness of these algorithms. However, this increases the number of training iterations and the time required or reduces the efficiency at times .

REFERENCES

- [1] R. Kafieh et al., "COVID-19 in Iran: Forecasting a Pandemic Using Deep Learning", Computational and Mathematical Methods in Medicine, vol. 2021, pp. 1-16, 2021.
- [2] AZ Khuzani, M. Heidari and A. Shariati, "COVID-Classifer: An automated machine learning model to assist in the diagnosis of COVID-19 infection in chest x-ray images", Scientific Reports, vol. 11, p. 1-6, 2021
- [3] M. Kukar et al., "COVID-19 diagnosis by routine blood tests using machine learning", 2020.
- [4] Y. Zoabi and N. Shomron, "COVID-19 diagnosis prediction by symptoms of tested individuals: a machine learning approach", MedRxiv, 2020.
- [5] VT Varna and P. Husbands, "Genetic Algorithm Assisted HIDMS-PSO: A New Hybrid Algorithm for Global Optimization," 2021 IEEE Congress on Evolutionary Computation (CEC), June 28 - July 1, 2021, DOI: 10.1109/CEC45853.2021.9504852.
- [6] A. Raghavan, P. Maan, and AKB Shenoy, "Optimization of Day-Ahead Energy Storage System Scheduling in Microgrid Using Genetic Algorithm and Particle Swarm Optimization," IEEE Access, vol. 8, 2020, DOI: 10.1109/ACCESS.2020.3025673.

- [7] DR Ramdania, M. Irfan, F. Alfarisi, and D. Nuraiman, "Comparison of genetic algorithms and Particle Swarm Optimization (PSO) algorithms in course scheduling," in 4th Annual Applied Science and Engineering Conference Journal of Physics: Conference Series, 2019, doi: 10.1088/1742-6596/1402/2/022079.
- [8] MU Siregar, I. Setiawan, NZ Akmal, D. Wardani, Y. Yunitasari, and A. Wijayanto, "Optimized Random Forest Classifier Based on Genetic Algorithm for Heart Failure Prediction," in 2022 Seventh International Conference on Informatics and Computing (ICIC), 2020, doi: 10.1109/ICIC56845.2022.10006987.
- [9] AN Fauziyah and WF Mahmudy, "Hybrid Genetic Algorithm for Optimization of Food Composition on Hypertensive Patient," International Journal of Electrical and Computer Engineering (IJECE), vol. 8, no. 6, pp. 4673-4683, December 2018, ISSN: 2088-8708, doi: 10.11591/ijece.v8i6.pp4673-4683.
- [10] A. Adriyendi and Y. Melia, "Optimization using Genetic Algorithm in Food Composition," International Journal of Computing and Digital Systems, vol. 10, no. 1, Oct. 2021, ISSN: 2210-142X, doi: 10.12785/ijcds/100191.
- [11] Farzin S, John A., 2016. "Expert System for Determining Discharge JB Matias, EM Porras, and AC Fajardo, "A hybrid particle swarm optimization to solve dietary menu planning," Journal of Critical Reviews, vol. 07, issue 15, 2020, ISSN: 2394-5125.
- [12] F. Omidinasab and V. Goodarzimehr, "A hybrid particle swarm optimization and genetic algorithm for truss structures with discrete variables," Journal of Applied and Computational Mechanics, vol. 6, no. 3, pp. 593-604, 2020, doi: 10.22055/JACM.2019.28992.1531.
- [13] T. Suratno, N. Rarasati, GZ, "Optimization of Genetic Algorithm for Implementation Designing and Modeling in Academic Scheduling," EKSAKTA, vol. 20, no. 1, pp. 166-174, April 2019. DOI: 10.24036/eksakta/vol20-iss01/166
- [14] LT Utomo, F. Wahyudi, "Implementation Of Genetic Algorithm In Optimization Of Subject Scheduling," JEEMECs (Journal of Electrical Engineering, Mechatronics and Computer Science), vol. 3, no. 2, pp. 125-134, August 2020. DOI: 10.26905/jeemecs.v3i2.4346
- [15] KT Hidayat, R. Arifudin, A. Alamsyah, "Genetic Algorithm for Relational Database Optimization in Reducing Query Execution Time," Scientific Journal of Informatics, vol. 5, no. 1, pp. 1-10, May 2018.
- [16] M. Cai, "An Improved Particle Swarm Optimization Algorithm and Its Application to the Extreme Value Optimization Problem of Multivariable Function," Hindawi Computational Intelligence and Neuroscience, vol. 2022, Article ID 1935272, 9 pages, 2022. DOI: 10.1155/2022/1935272
- [17] J. Li, Y. Sun, S. Hou, "Particle Swarm Optimization Algorithm with Multiple Phases for Solving Continuous Optimization Problems," Hindawi Discrete Dynamics in Nature and Society, vol. 2021, Article ID 8378579, 13 pages, 2021. DOI: 10.1155/2021/8378579
- [18] S. Chaudhary, VK Sharma, RN Thakur, A. Rath, P. Kumar, S. Sharma, "Modified Particle Swarm Optimization Based on Aging Leaders and Challengers Model for Task Scheduling in Cloud Computing," Hindawi Mathematical Problems in Engineering, vol. 2023, Article ID 3916735, 11 pages, 2023. DOI: 10.1155/2023/3916735
- [19] A. Zhang, Z. Liang, "Energy efficiency optimization of ships based on particle swarm optimization," E3S Web of Conferences, vol. 375, 03016, 2023. DOI: 10.1051/e3sconf/202337503016
- [20] Song, Y.; Liu, Y.; Chen, H.; Deng, W. A Multi-Strategy Adaptive Particle Swarm Optimization Algorithm for Solving Optimization Problems. Electronics 2023, 12, 491. <https://doi.org/10.3390/electronics12030491>
- [21] Dya, T., Blaise, BB, Betchewe, G. and Alidou, M. (2021) Implementation of Particle Swarm Optimization Algorithm in Matlab Code for Hyperelastic Characterization. World Journal of Mechanics, 11, 146-163. <https://doi.org/10.4236/wjm.2021.117011>